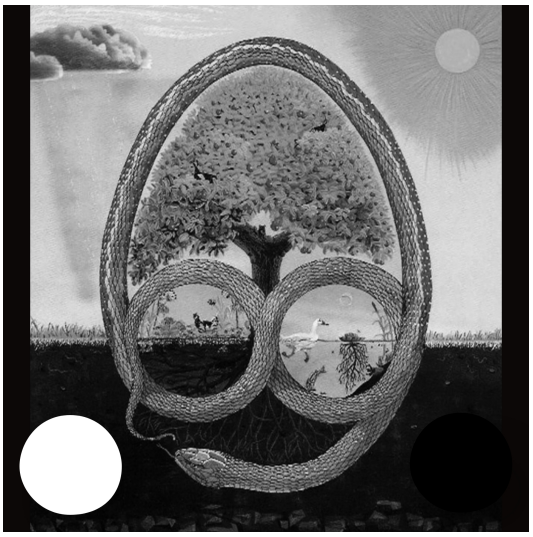

Unturf Automated General Intelligence: Merkle Providence Reverse RAG

Provenance-Preserving Cache Chains for Small Language Models

How Merkle trees turn Reverse RAG into a verifiable, shared knowledge layer.

How Hermes 3 Llama 3.1 8B answers with the confidence of a verified record, not a guess.

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Abstract

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Reverse Retrieval Augmented Generation (Reverse RAG) demonstrated that client-side context injection lets small 8B-parameter language models outperform much larger models on page-specific questions. Our reference implementation runs **Hermes 3**, a NousResearch instruction fine-tune of Meta's Llama 3.1 8B, served freely at uncloseai.com. Our prior work showed that the client already holds the document. No vector database required. No embedding pipeline. No retrieval failures.

Merkle Providence Reverse RAG extends this with a second insight: the client not only holds the document, it can *remember what it already computed about that document*, prove that memory came from an unmodified source, & share that verified knowledge with others on public or private chains.

A Merkle tree over document content chunks produces a root hash: a 32-byte fingerprint that changes when any part of the document changes. Pairing this root hash with a question hash yields a cache key. A cache hit means a previously computed answer returns instantly, with a Merkle proof of origin. A cache miss triggers fresh inference, stores the result, & extends our chain.

Unfirehose serves as our chain substrate. Every inference session produces JSONL records that feed into SQLite at `~/.unfirehose/unfirehose.db`. Public chains broadcast verified answer records to peers. Private chains stay local. Either way, a small model gains access to a growing body of pre-verified answers, each carrying cryptographic proof of the document version that produced it.

This combination removes two limits that have constrained small models:

1. **The repetition tax**: answering the same question about the same document on every visit, burning inference tokens & time, producing answers that may vary slightly each run.
2. **The trust deficit**: small models hallucinate. Cached answers with Merkle proofs do not hallucinate. They either match a verified record or they do not exist yet.

Once published, this technique forces a reckoning: every RAG system that cannot prove provenance of its retrieved chunks operates on unverifiable context. Merkle Providence makes provenance a first-class primitive.

We call this stack **Web 2.5**: the existing web's documents & URIs, augmented with machine learning context injection & Merkle-verified answer chains, without requiring a blockchain, a new protocol, or replacement of any existing infrastructure. Web 2.5 runs on top of what already exists. Any page. Any browser. Any small model. Public or private chain. No permission required.

Reference Model: Hermes 3 Llama 3.1 8B

Our reference implementation runs on **Hermes 3**, a fine-tune of Meta's Llama 3.1 8B developed by NousResearch. Hermes 3 builds on Llama 3.1's 128k context window & adds instruction alignment, function-calling discipline, & reasoning consistency that raw base models lack. NousResearch trained Hermes 3 to follow multi-turn conversation formats precisely, making it reliable for our structured providence cache interactions.

We chose Hermes 3 Llama 3.1 8B for three reasons:

Size matches our thesis. 8 billion parameters represent a model any person can run on consumer hardware. Our reference deployment hardware & quantization details stay current at uncloseai.com/inference.html. Our technique produces no value if it requires expensive hardware to demonstrate.

Fine-tuning demonstrates our point. Hermes 3 shows that fine-tuning a base model for instruction following, not scaling parameters, produces the reliability gains that matter for user-facing applications. Our Merkle Providence layer extends this logic: fine-tuned behavior + verified cached context beats raw parameter count.

uncloseai.com serves it freely. Our public inference endpoint at uncloseai.com runs Hermes 3 Llama 3.1 8B at no cost to any user. Our Reverse RAG pipeline in `uncloseai.js` targets this endpoint by default. Our Merkle Providence cache layer builds directly on this existing infrastructure. Any person anywhere, on any device with a browser, accesses our same Hermes 3 instance our paper describes.

Hermes 3 Llama 3.1 8B outperforms many larger models on page-specific questions when Reverse RAG supplies full context. With Merkle Providence caching, it delivers consistent, verifiable answers across sessions. Our paper uses Hermes 3 throughout all examples & benchmarks.

1. The Problem Reverse RAG Left Open

Reverse RAG solved context retrieval. Client-side DOM extraction replaced vector databases. Full-page injection replaced chunk fragmentation. Small models with perfect context outperformed large models with partial context.

One problem remained: **repetition**.

A user visits a technical document. Hermes 3 Llama 3.1 8B reads 8,000 tokens of page content, classifies the page, & answers their question. One hour later, a different user visits our same document. Our same Hermes 3 instance reads our same 8,000 tokens, classifies our same page, & answers a similar question again. Tomorrow, a third user. Our same 8,000 tokens. Our same classification. Our same inference cycle.

Nothing remembered. Nothing reused. No proof that any two answers came from our same source.

This repetition carries three costs:

- **Compute cost**: every visit re-runs inference on content that has not changed.

- **Consistency cost**: stochastic inference on identical inputs produces slightly different answers. Two users asking near-identical questions may receive meaningfully different responses from our same page.
- **Trust cost**: no mechanism exists to verify that an answer actually derived from a specific document version. A model can claim anything.

Merkle Providence addresses all three.

2. Merkle Trees as Content-Addressed Cache Keys

A Merkle tree splits a document into chunks, hashes each chunk, then builds a binary tree of hashes upward until a single root hash remains. This root hash has two properties that make it ideal for caching:

Property 1: Determinism. Our same document always produces our same root hash. Two clients processing our same page independently compute identical roots. No coordination required.

Property 2: Tamper sensitivity. Any change to any byte of the document changes at least one leaf hash, which propagates upward to invalidate our root. A document that changed since our last visit produces a new root hash automatically. Our cache never serves a stale answer for a modified document.

Cache key construction:

```
document_root = merkle_root(chunks(page_content))
question_key  = sha256(normalize(user_question))
cache_key     = document_root + ":" + question_key
```

A cache hit at this key means: *an earlier session computed an answer to this question, about this exact version of this document.* Our answer arrives in O(1) time. Our Merkle proof confirms our document version.

A cache miss means: run inference, store our result, extend our chain.

Git & Mercurial are already Merkle trees. For source code repositories, we do not need to build our own Merkle tree. Every git commit hash IS a Merkle root: a SHA-1 (or SHA-256 in git 2.x) digest of our entire repository tree at that point in time. Every hg commit carries an identical guarantee. The version control system has already done our hashing work.

For codebases, our cache key simplifies to:

```
document_root = git_commit_hash   # or hg changeset hash
question_key  = sha256(normalize(user_question))
cache_key     = document_root + ":" + question_key
```

When a developer commits, our cache key changes automatically. Prior answers expire. Fresh inference runs against our new commit. No manual invalidation. No polling. Our version control system signals our cache boundary by design.

3. The Providence Layer: Proof of Origin

"Providence" carries two meanings that both apply here:

Provenance: where something came from. A Merkle proof lets any verifier confirm that an answer derived from a specific document version without receiving our full document. Our proof path, a sequence of sibling hashes from leaf to root, constitutes a compact cryptographic certificate of origin.

Foresight: the system appears to "already know" answers to questions asked before. From a user's perspective, a cache hit looks like instant comprehension. Our model responded before seemingly finishing to read. This resembles foresight because our prior sessions did our work ahead of time.

A providence record for a cached answer contains the full inference context:

```
{
  "cache_key":        "abc123...:def456...",
  "document_root":    "sha256:abc123...",
  "document_uri":     "https://example.com/docs/api",
  "question_hash":    "def456...",
  "question_text":    "What does the rate_limit field mean?",
  "answer_text":      "The rate_limit field specifies...",
  "merkle_proof":     ["sha256:111...", "sha256:222...", ...],
  "model_id":         "adamo1139/Hermes-3-Llama-3.1-8B-FP8-Dynamic",
  "base_uri":         "https://hermes.ai.unturf.com/v1",
  "temperature":      0.1,
  "source_type":      "web",
  "git_commit":       null,
  "conversation_hash": "9a3f...",
  "turn_number":      1,
  "created_at":       1744000000,
  "hit_count":        0
}
```

Our providence record collects maximum metadata, but only a subset enters our cache key. This distinction matters.

Tier 1 — Cache Key Inputs. A difference in any of these fields produces a distinct cache entry:

- `document_root` — content fingerprint (Merkle hash or git/hg commit)
- `question_hash` — SHA-256 of normalized question text. Four-step canonicalization, in order: NFC + whitespace collapse + edge-strip, then case-fold to lowercase, then trailing-punctuation strip, then drop standalone English articles (`the` / `a` / `an`). Trailing chars in the strip set today: `.?!,;:` (ASCII), ■■■■ (CJK full-width), … (ellipsis). Pairs (`" ' ) ] }`) are NOT stripped — naive one-sided stripping breaks balance, and apostrophes carry meaning (`X's` ≠ `X`). Articles are matched as exact lowercase standalone tokens, so substrings like `thesis` (contains `the`) stay untouched. Equivalence class: `who is X`, `who is X?`, `Who Is X.`, `who is the X`, `who is a X`, `who is an X■` all map to the same `question_hash`, so a providence_cache hit returns the same answer for any of those forms.
- `model_id` — canonical model identifier
- `model_revision` — exact weights revision (HuggingFace commit hash or equivalent)
- `quantization` — fp16, bf16, fp8, int8, int4, q4_k_m, etc. Same weights at different quantizations produce measurably different answers.
- `conversation_hash` — SHA-256 of the full normalized industry-standard chat messages array: every turn in order, system prompt through the final user message. A single-turn Q&A and a 6-turn conversation that arrives at the same final question produce different answers & different cache keys. Callers hash client-side & send only the hash. Message content never enters our logs — Operation Voyeur.
- `seed` — RNG seed when set. A seeded run produces a deterministic answer; a different seed produces a different one. Null seed entries cache on the other five fields alone.

Tier 2 — Metadata Only. Stored for research, routing, & audit. Not part of the key:

- `base_uri` — inference endpoint URI
- `temperature`, `top_p`, `top_k` — sampling parameters
- `repetition_penalty`, `frequency_penalty`, `presence_penalty`
- `max_tokens`, `context_window`
- `backend` — vllm, llama.cpp, ollama, transformers, TGI
- `node_id` — mesh node that served the request
- `inference_ms` — wall-clock inference time

Tier 2 metadata does not affect cache correctness. Two answers produced at different temperatures from the same model, revision, & quantization, answering the same question about the same document, represent the same logical answer — stochastically equivalent. We store the temperature of the first run that populated our cache. Future hits return that answer regardless of the caller's temperature setting.

Any recipient of this record can verify our Merkle proof against our document root. If our document root matches what they compute from our same URI, our answer carries full provenance. If their computation produces a different root, our document changed, our answer no longer applies to their version.

4. Unfirehose as Chain Substrate

Unfirehose collects JSONL session data from machine learning harnesses across a mesh of compute nodes. Every tool call, every model response, every session event flows into `~/.unfirehose/unfirehose.db` via our background ingestion worker.

Our Merkle providence layer extends this existing infrastructure with a providence cache table:

```
CREATE TABLE providence_cache (
  cache_key        TEXT PRIMARY KEY,
  document_root    TEXT NOT NULL,
  document_uri     TEXT NOT NULL,
  question_hash    TEXT NOT NULL,
  question_text    TEXT NOT NULL,
  answer_text      TEXT NOT NULL,
  merkle_proof     TEXT NOT NULL,  -- JSON array of sibling hashes
  model            TEXT NOT NULL,
  created_at       INTEGER NOT NULL,
  chain            TEXT NOT NULL DEFAULT 'private',
  peer_count       INTEGER NOT NULL DEFAULT 0
);

CREATE INDEX idx_providence_root ON providence_cache(document_root);
CREATE INDEX idx_providence_uri  ON providence_cache(document_uri);
```

Two chain modes:

Private chain: records stay local at `~/.unfirehose/`. Our personal session history. Our model learns from our own prior work on documents we visit repeatedly. Our chain belongs to us alone.

Public chain: records broadcast to peers via our unfirehose router daemon. Any node on our mesh that has processed our same document root can contribute answers. Any node that queries our same document root can receive pre-verified answers from others. Our chain grows with every participant.

Public chains do not share document content. They share proofs. Our Merkle proof proves an answer came from a specific document version without transmitting our document. Document content stays on our originating client, exactly as Reverse RAG intended.

5. How Small Models Punch Outside Their Weight Class

Parameter count governs a model's ability to reason from first principles. Context quality governs a model's ability to answer questions about specific documents. Verified memory governs a model's ability to give consistent, trustworthy answers over time.

Large models win on reasoning from first principles. Small models, given the right tools, win on everything else.

Merkle Providence Reverse RAG gives small models three tools:

Tool 1: Full context (from Reverse RAG). Our same page injection described in our prior paper. Our model receives every word, every heading, every entity, every code block of our document. Nothing retrieved by cosine similarity. Nothing fragmented. Our full document, live from our DOM.

Tool 2: Verified recall (from our providence cache). Before our model reads our document, we check our cache. A cache hit returns our answer in milliseconds with a Merkle proof. Our model delivers a verified

answer faster than any large model can load our full context. Our small model "already knew" because our prior session did our work.

Tool 3: Peer-verified answers (from public chains). On a public chain, our small model benefits from every session any participant ran against our same document version. A document processed by 100 sessions across our mesh produces a rich cache. Our Hermes 3 Llama 3.1 8B instance answers from our collective record, not just its own prior work.

These three tools operate at different timescales:

- **O(1)**: cache hit, local or peer. Instant. No inference required.
- **O(context)**: Reverse RAG with full page injection. Fast. One inference pass over complete context.
- **O(reasoning)**: fresh inference on novel questions. Standard. One inference pass, uncached.

A large model operates exclusively at O(reasoning). Our small model with Merkle Providence operates at O(1) for previously answered questions, O(context) for new questions on known documents, & O(reasoning) only for truly novel questions on new documents.

6. Architecture

Stage 1: Document Fingerprinting

Our same DOM extraction from Reverse RAG runs first. Once our extraction completes, we split our content into 512-token chunks & build our Merkle tree:

```
function merkleRoot(chunks) {
    let layer = chunks.map(c => sha256(c));
    while (layer.length > 1) {
        const next = [];
        for (let i = 0; i < layer.length; i += 2) {
            const left  = layer[i];
            const right = layer[i + 1] ?? left;  // odd node doubles itself
            next.push(sha256(left + right));
        }
        layer = next;
    }
    return layer[0];
}
```

Our root hash serves as our document fingerprint for our entire session. One hash computation. No server calls.

Stage 2: Cache Lookup

Before any LLM call, we check our providence cache:

```
async function lookupProvidence(documentRoot, questionText) {
    const questionHash = sha256(normalize(questionText));
    const cacheKey     = `${documentRoot}:${questionHash}`;

    // Check local chain first
    const local = await db.get(
        'SELECT * FROM providence_cache WHERE cache_key = ?',
        [cacheKey]
    );
    if (local) return { answer: local.answer_text, proof: local.merkle_proof, source: 'local' };

    // Check public chain peers
    if (publicChainEnabled) {
        const peer = await queryPeers(cacheKey);
        if (peer && verifyMerkleProof(peer.merkle_proof, documentRoot)) {
            await storeLocal(peer);  // cache locally for future hits
```

```
            return { answer: peer.answer_text, proof: peer.merkle_proof, source: 'peer' };
        }
    }

    return null;  // cache miss, proceed to inference
}
```

Stage 3: Inference & Cache Write

On a cache miss, our standard Reverse RAG inference runs. Once our answer arrives, we store our providence record:

```
async function storeProvidence(documentRoot, documentUri, question, answer, model) {
    const questionHash = sha256(normalize(question));
    const cacheKey     = `${documentRoot}:${questionHash}`;
    const merkleProof  = computeProof(documentRoot, question);  // path to root

    await db.run(`
        INSERT OR REPLACE INTO providence_cache
        (cache_key, document_root, document_uri, question_hash, question_text,
         answer_text, merkle_proof, model, created_at, chain)
        VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)
    `, [cacheKey, documentRoot, documentUri, questionHash, question,
        answer, JSON.stringify(merkleProof), model, Date.now(), chainMode]);

    if (chainMode === 'public') broadcastToMesh(cacheKey);
}
```

Stage 4: Proof Verification

Any party that receives a providence record can verify it:

```
function verifyMerkleProof(proof, claimedRoot, leafContent) {
    let current = sha256(leafContent);
    for (const sibling of proof) {
        current = sha256(current < sibling
            ? current + sibling
            : sibling + current);
    }
    return current === claimedRoot;
}
```

If our document root computed from our verifier's live DOM matches `claimedRoot`, our answer carries full provenance. Our answer derives from our exact document version our verifier currently views. If our roots diverge, our document changed. Our cache entry no longer applies.

7. Public vs. Private Chains

Property	Private Chain	Public Chain
Storage	`~/.unfirehose/` local only	Broadcast via unfirehose router mesh
Participants	One user, one machine	Any node on our mesh
Document content shared	Never	Never (proofs only)
Answer content shared	Never	Yes, with Merkle proof
Trust model	Self-trust: our own prior work	Peer-trust: verified by Merkle proof
Best for	Personal document libraries, sensitive internal docs	Open documentation, public research, shared reference material

Cache growth rate	Linear in our own sessions	Linear in mesh participant sessions

Private chains suit corporate knowledge bases, personal research notes, any document our user does not want to share. Our local model benefits from our own prior sessions. No data leaves our machine.

Public chains suit open documentation, public APIs, research papers, any document where sharing verified answers benefits our community. No document content leaves our originating client. Proofs travel instead.

In our proposed design, our unfirehose router daemon would manage chain membership & peer discovery. Our user opts in to public chains per-document-domain or per-session. Our current implementation stays private.

8. Forcing the Issue

Some technical formulations compel structural change not because they require adoption but because their existence makes our prior approach indefensible.

Our Merkle Providence layer forces one question onto every RAG system: *can you prove where your context came from?*

Traditional RAG retrieves chunks by cosine similarity. No chunk carries a Merkle proof. No retrieval carries a document fingerprint. Our retrieved context could come from a stale index, a corrupted chunk, or a document that changed since our last ingestion. Our RAG pipeline trusts our embedding similarity as a proxy for relevance. Nothing verifies our source.

Once Merkle-proved answers exist in our ecosystem, unverified RAG answers carry an implicit disclaimer: *I retrieved this from somewhere, I believe it matches your question, but I cannot prove our source was our document you currently view.*

Our providence cache answers carry a different posture: *I computed this answer from our exact bytes of our document whose root hash I can show you. Verify it yourself.*

This posture change does not require every system to adopt Merkle trees immediately. It requires every system to acknowledge that provenance matters, & that systems which cannot demonstrate provenance operate at a disadvantage in trust-sensitive contexts.

Medical documentation. Legal reference material. Security advisories. Financial disclosures. Any domain where "I retrieved something similar" falls short of "I can prove this came from our exact source you cited."

9. What the Providence Cache Curates & Aggregates

Every inference pass through our Merkle Providence layer produces a verified record. Over time these records accumulate into a structured, queryable knowledge base. Here we describe what our cache collects & what emerges from that collection.

Document fingerprints. Every document processed — web page, git-versioned file, or raw content — contributes a `document_root`: a deterministic hash of its content at a specific moment. Our cache accumulates a catalog of every document any session has processed, keyed by content not by name or URI. A document that moves keeps its fingerprint. A document that changes gets a new one.

Answer records. For every (document, conversation, model, revision, quantization) combination, our cache stores one verified answer. The answer record carries: the question text, the answer text, the full Merkle proof path, the conversation hash that produced the context, & the model identity & inference configuration that produced the output. These records aggregate into a corpus of verified knowledge about our document collection.

Conversation context fingerprints. Our `conversation_hash` field captures the full industry-standard chat messages array, system prompt, all prior turns, & the final user message, as a single SHA-256. Our cache accumulates not just answers but the exact conversational paths that produced them. Two agents that reached the same question via different conversation histories leave distinct fingerprints. Over time we observe which conversational approaches yield cache hits & which always miss — signal for prompt engineering.

Hit counts & reuse rates. Every cache lookup that finds an existing record increments `hit_count` & updates `last_hit_at`. Our cache aggregates these into a measure of knowledge reuse: which documents get queried repeatedly, which questions recur, which model + quantization combinations serve the most requests. High-hit records represent our most valuable verified knowledge — answers our mesh has found useful enough to retrieve many times.

Model & endpoint diversity. Each record carries `model_id`, `model_revision`, `quantization`, & `base_uri`. Our cache aggregates a record of which models answered which questions about which documents. Over time this produces a comparative dataset: the same question about the same document answered by Hermes 3 at fp8, at int4, by a different model entirely. Our cache does not collapse these — it preserves the full diversity for analysis.

Inference economics. `inference_ms`, `backend`, `node_id`, & sampling parameters aggregate into a routing intelligence layer. Our cache knows which nodes answer quickly, which backends run which models efficiently, & which quantization levels trade answer quality for speed. Cache hits carry this history forward — a future routing decision can prefer a node whose cached answers already proved useful.

Git-versioned knowledge graphs. For codebase queries, our `git_commit` field anchors every answer to an exact repository state. Our cache accumulates a version-annotated knowledge graph of our codebase: "at commit abc123, this function does X; at commit def456 it does Y." When a developer asks "what changed?", our cache surfaces the delta between two commit-keyed answer sets without re-reading any file.

9b. Unfirehose Integration (Proposed)

Unfirehose collects JSONL from Claude Code, Orchestra, uncloseai, & agnt.gg harnesses across our mesh. Our providence cache layer does not yet exist in production. This section describes our proposed integration.

We propose extending unfirehose with a `providence_cache` table & three new API routes:

`GET /api/providence?uri=...`

Returns all cached answer records for a document URI. Includes document root hash, question hashes, answer previews, & peer counts. Lets users audit what our mesh knows about a document.

`POST /api/providence`

Stores a new providence record. Accepts `{ document_root, document_uri, question_text, answer_text, merkle_proof, model }`. Validates our proof before storing.

`GET /api/providence/peers?root=...`

Queries our mesh for cached records matching a document root hash. Peer querying remains a future capability, opt-in per domain. Our initial implementation stays private: local cache only, no mesh broadcasting.

Our existing unfirehose dashboard would gain a Providence page: a view of our cache landscape across documents, cache hit rates per domain, & chain health.

10. Relationship to Zero-Knowledge Proofs

A Merkle proof constitutes a specific form of zero-knowledge argument: *I can prove this leaf exists in this tree without revealing our other leaves.*

In our context: *I can prove this answer came from this document version without revealing our full document.*

Our full ZKP apparatus (zk-SNARKs, Bulletproofs) provides stronger guarantees & more complex proofs. Merkle proofs provide a simpler, faster, & more practical subset: membership proofs sufficient for our provenance use case.

Our connection to ZKP deepens when our public chain handles sensitive documents. A user could prove their answer derived from a specific confidential document without revealing our document content, our question text, or our answer text. Only our proof path & our document root travel over our network. Our verifier confirms our proof without seeing our data.

This positions our public chain as opt-in verifiable knowledge sharing, not compelled disclosure.

11. Why This Matters for Permacomputer

Permacomputer infrastructure grows knowledge that outlasts our individual sessions, our individual machines, & our individual participants. Code seeds sprout on any abandoned technology. Knowledge should do our same.

A private chain that lives only on one machine dies with that machine. Our public chain on our unfirehose mesh survives node failures because our same document root on a different node still matches our cached proofs on our other nodes. Our knowledge persists as long as any mesh participant holds a copy.

Our Merkle structure aligns with permacomputer values:

Truth: proofs over assertions. Our answer carries a mathematical certificate, not just a model's confidence score.

Freedom: private chains for sensitive knowledge, public chains for shared knowledge. Our user chooses. No platform extracts our knowledge without consent.

Harmony: cache hits replace repeated inference. Our same compute produces more answers over time. Our mesh grows more efficient as our cache fills.

Love: small models with providence caches remove our infrastructure barrier. Hermes 3 Llama 3.1 8B on a consumer GPU, connected to our public chain, answers with our collective memory of our mesh. Our permacomputer does not require our most expensive hardware to deliver our most reliable answers.

12. Implementation Roadmap

Phase 1: Local private chain (unfirehose extension)

- Add `providence_cache` table to our unfirehose SQLite schema
- Extend our uncloseai.js Reverse RAG pipeline with pre-inference cache lookup & post-inference cache write
- Add `GET /api/providence` & `POST /api/providence` routes to our unfirehose web app
- Add Providence view to our dashboard

Phase 2: Public chain via unfirehose router

- Extend our router daemon with `/providence/broadcast` & `/providence/query` endpoints
- Add peer discovery to our mesh probe logic
- Add per-domain opt-in controls to our settings page
- Add peer contribution metrics to our Providence dashboard view

Phase 3: Proof verification in browser

- Implement `verifyMerkleProof()` in our uncloseai.js client
- Surface proof status in our chat UI (verified badge vs. fresh inference)
- Add document root display so users can audit our source fingerprint

Phase 4: Harness-level knowledge graph via unfirehose

Unfirehose collects JSONL from every harness session across our mesh. Every file read, every tool call, every assistant response produces a record. At the harness level, our Merkle Providence layer extends beyond web pages to entire codebases & research corpora:

- **Commit-keyed cache**: when a harness reads a file, we record `(git_commit_hash, file_path, question_hash)` $\rightarrow$ `answer`. Our same question about our

same file at our same commit hits our cache on every future session, on any node in our mesh, without re-reading the file.

- **Knowledge graph construction**: unfirehose builds a graph of concepts, files, functions, & sessions from harness JSONL. Nodes carry Merkle-verified summaries. Edges carry session provenance. Our graph grows with every harness run.
- **Graph-first traversal**: a harness navigating a large codebase queries our knowledge graph first. Only cache misses trigger fresh file reads & inference. Whether this reduces token usage, & by how much, remains for future measurement once a prototype exists.
- **Cross-session continuity**: a new harness session on our same codebase at our same commit inherits our full prior knowledge graph. No re-reading. No re-summarizing. Our mesh memory persists across sessions, machines, & agents.

This positions unfirehose not just as a log aggregator but as a shared, verifiable knowledge substrate for our entire mesh of machine learning agents.

13. Aborist: A Python Reference Implementation

Our uncloseai.js Reverse RAG client lives in our user's browser. Our Merkle Providence cache layer described above also belongs in our browser, & in a Next.js logger that aggregates JSONL across our mesh. A complementary surface exists at our other end of our pipeline: a server-side, content-addressed document store that ingests entire corpora at once, computes Merkle commitments at scale, & serves cached answers to any peer that produces matching cache keys.

Aborist (https://git.unturf.com/engineering/unturf/aborist) is our Python reference implementation of that surface. Our codebase ships under AGPL-3.0-only, runs against hermes.ai.unturf.com by default, & implements every primitive described in Sections 1 through 12 of this paper as working code with measurements on a real corpus.

13.1 Schema: v9.8 Admissibility Ledger

Aborist's providence cache uses an 8-dimensional cache key, a strict superset of our 6-dimensional key in Section 3. Two extra dimensions formalize what was previously implicit:

```
cache_key = source_root | question_hash | model_profile_hash
          | conversation_hash | governance_policy_hash
          | schema_version | canonicalization_version
          | chunking_version
```

Bumping any one stales every prior cache record, so a chunker change or a canonicalization rule change cleanly partitions our cache namespace. Records carry a `falsification_state ∈ {live, failed, stale, quarantined}`: queries always filter on `state='live'`, & a separate `falsifications` table logs every state transition with `cache_key`, reason, actor, & `audit_event_hash`.

Every state-changing operation (ingest, distill, falsify, evict, rehydrate, burn, mesh epoch rotation, snapshot creation) appends one row to a tamper-evident audit chain:

```
event_hash = sha256(prev_event_hash || canonical_json(body))
```

Verification is purely local: re-hash forward & confirm. A break in our chain signals tampering on our DB at a specific event index.

Two leaf-removal verbs cover different stages of corpus life:

- **Falsify** (audit-preserving). Flips `falsification_state` from `live` to `failed` / `stale` / `quarantined` and writes a row to `falsifications` (cache_key, reason, actor, timestamp, audit_event_hash). Cache lookups always filter on `state='live'` so the record stops being

reachable, but it stays on disk as forensic history. Use when downstream consumers may have ingested the answer.

- **Burn** (kindergarten leaf delete). Actually deletes the row. Kept off the default verb path because it forfeits history. Refuses if the leaf has children — for `providence_cache` leaves: any falsifications referencing the cache_key; for `documents`: any derivations using the doc as a source, any incoming edges, any providence records keyed on its document_root; for `cores`: same as documents plus downstream cores in derivations. `--force` overrides for cleanup edge cases & records `forced=True` plus the child count at burn time in the audit body. Always writes a `providence_burn` / `document_burn` / `core_burn` event regardless of forcing. Use during early/scratch corpus building before downstream consumers exist; otherwise prefer falsify.

Both surfaces (`aborist providence --falsify` & `aborist burn`) compose with cross-shard scope. Chain integrity after either operation is verifiable in O(1) per shard via `make chain-check-shards` — a `LEFT JOIN` query that counts dangling `prev_event_hash` references; 0 = chain intact, non-zero = a break to investigate.

13.2 Multi-Source ETL

Aborist ships with five source adapters, each yielding *Document* objects through our same *ingest_source* pipeline:

- **Wikipedia Phase III SQL dumps** (2003–2005). Hand-rolled streaming parser for MySQL extended INSERT syntax, bz2-aware, ~3 minutes per 128k articles on commodity hardware.
- **Wikipedia Phase IV XML dumps** (2006+). Streaming *iterparse* over *pages-articles.xml.bz2* with bounded memory; supports *pages-meta-history.xml.bz2* for full revision history. Tested against 2010-11 enwiki at 6.2 GB compressed.
- **xAI Grok account exports**. One Document per conversation, every message rendered with sender/model/timestamp inline.
- **Git & Mercurial repos** (self-play). Walks *git ls-tree HEAD* or *hg manifest*; one Document per text file. Re-ingest after new commits auto-chains via *supersedes* edges, so our Merkle tree grows alongside our repository.
- **HTML pages**. Robots-aware fetch with *selectolax* extraction; hyperlinks become outbound edges.
- **Live website crawl** (off by default, `aborist[crawler]` extras). BFS-discovers same-domain URLs from a seed via a verbatim lift of agents.ai.unturf.com's *AsyncWebFetcher* (ethical: respects robots.txt Disallow + Crawl-delay, with a *fast_mode* flag that drops the delay for trusted hosts). Discovered URLs are filtered: feeds (atom/rss/rdf/sitemap) and image/video/audio paths are crawl infrastructure, not knowledge content, & skip ingest. ETag and Last-Modified per response are captured into `document_http_meta` so a subsequent `aborist crawler recrawl-check` round can issue conditional `If-None-Match` / `If-Modified-Since` HEAD requests; 304 = fresh (no body transfer), 200 = stale (re-ingest), 404/410 = gone, other = unreachable. The crawl shard filename is derived from the seed hostname (`crawl_<host_with_underscores>.db` under `$(SHARDS_DIR)`) so cross-shard query picks up new domains without configuration.

Adding a seventh corpus is one new *Source* subclass. Our contract is intentionally minimal: yield Documents, our pipeline canonicalizes, chunks, hashes, persists.

13.3 Storage Cheats: 67% Reduction Measured

Three orthogonal optimizations stack against SQLite-resident corpora:

- **Zstandard-compressed chunk content.** Plaintext bodies compress 3 to 5x at level 3. Magic-byte detection on read makes legacy uncompressed rows transparent. Cores stay plaintext so SQL substring queries continue working.

- **WITHOUT ROWID on edges.** Our composite primary key covers every column, so a default rowid-keyed table near-doubles row data in our PK index. WITHOUT ROWID makes our table itself a B-tree keyed on our PK. We dropped our partial dst_uri index too, since only one analytic query actually filters on dst_uri.
- **Contentless FTS5.** Our prior schema had FTS5 store its own copy of every chunk's text. Contentless mode (*content=''*, *contentless_delete=1*) keeps only our inverted index. Snippets are built in Python by joining *chunks_fts.rowid* = *chunks.chunk_id* & decompressing. Requires SQLite 3.43+.

Apples-to-apples reingest of our 2003-05-16 enwiki cur dump (128,245 documents):

```
OLD schema:  2.6 GB across 4 shards    (~21 KB / doc)
NEW schema:  857 MB across 4 shards    (~6.7 KB / doc)
reduction:   -67%
```

Same document set, same Merkle roots, identical query results. Storage cost dropped by two-thirds with no change to our content-addressed identity.

13.4 Retrieval Quality: Recall Fixes

Our reverse-RAG client picks one document; aborist picks K from a corpus & merges. The ranking pipeline grew through five live-corpus catches; each fix earned its place by failing on a real query. The current shape is a multi-stage filter with three accept paths & a stem-tolerant token matcher.

13.4.1 Title-search baseline dominated FTS5

A single-token title overlap against a generic word like `"intel"` used to score 60+, swamping body-relevant FTS5 hits scoring 5-15. Result: every `Intel_*`-prefixed legacy article flooded our top-K for any "intel" query, even when a contemporary article like `Pentium_4` had body relevance & no title overlap. Our fix dropped our title-search baseline so title overlap competes with FTS5 instead of monopolizing it. Title contributes a token-overlap boost (`overlap × 10.0`) on top of BM25; it doesn't run as a separate ranking signal.

13.4.2 BM25 short-doc bias surfaced stubs over topical articles

A 1-paragraph stub mentioning every query token once outranked a 30-page topical article where our same tokens are diluted by length. We added a body-coverage rerank using `sum(sqrt(count(t))) * 0.6` over query tokens for each surviving candidate. Sqrt scaling differentiates stubs from topical articles without letting enumerative list pages run away with our score.

Verified on our 2010-11 enwiki corpus (3.47M documents, no distill cores):

```
before:  Intel_4040 (1974 microcontroller stub) ranked #1
         for "what is the fastest intel CPU?"
after:   Intel_Core_2 #1, Intel_Atom #2, Intel_4040 #3
         answer cites Core i7 specifications correctly
```

13.4.3 Three accept paths in the title-relevance filter

A hit clears `_filter_by_title_relevance` if any one of three accept paths fires:

1. **Title-token overlap.** Strongest signal. Direct stem-aware match between the query's title-tokens and the document's title-tokens. Synonym fallback (`concepts.synonym_expand`) only fires for 1-token queries — multi-token queries that admit synonyms over-recall (an "amd"-querying agent picking up *Intel_** titles via the Intel/AMD rivalry group).
2. **TF-IDF core keyword overlap.** Closes the gap for neologisms that never appear in titles but are distinctive enough to be the TF-IDF top-keywords of conversation bodies. `"permacomputer"` was the canonical case — never a Wikipedia title, but a strong TF-IDF signal in Grok-conversation cores. Without this path, neologism queries returned empty.

3. **Body density.** `_body_density_passes` admits docs that mention at least `breadth_threshold` distinct query tokens AND have `total_mentions >= 3`. Cheap proxy for "actually about the topic" when neither title nor core caught it.

Rivalry exclusion (`rivalry_excluded` in `aborist/qa/concepts.py`) applies on every accept path: an Intel-titled doc dropped from an AMD query, & vice versa. Caught the case of competitor docs leaking into single-side queries when both share a topic page.

13.4.4 Breadth-required filter — multi-token queries demand co-occurrence

Pre-2026-04-29 the body-density rule was "at least HALF the tokens present" — too lenient for 2-token queries. Fox caught this on 2026-04-29: `"who is supermans girlfriend?"` admitted seven unrelated `Girlfriends`-titled articles (TV show, films, songs) because they had `girlfriend` in body but no `superman` whatsoever; the half-rule passed at 1-of-2.

The fix scales the breadth threshold with query length:

```
≤ 2 query tokens   require ALL distinct tokens present (in body and in title)
3+ query tokens    require N - 1 (one weak signal token may miss)
```

Plus the depth check (`total_mentions >= 3`) still gates body-density. Title-token-overlap shares the same scaling. The "3+ require N-1" path lets one query token miss for long queries where models throw filler in (`"what is the plot of the matrix film?"` admits docs that have all of `plot`/`matrix`/`film` even when `what` is filtered out as a stopword & `the` doesn't count).

Verified on the same enwiki corpus:

```
before:  Girlfriends (1991 album), Girlfriend (Avril Lavigne song),
         Girlfriends (TV series), Girlfriends Films, ...
         — 7 of 8 top hits unrelated to Superman
after:   Lois_Lane #1, Superman_in_film #2, Smallville #3, ...
         — Superman/Lois Lane mythology surfaces correctly
```

13.4.5 Stem-aware token matching — possessive & plural collapse

Same `supermans girlfriend` defect had a second cause: `superman's` from the question canonicalized to `supermans` (apostrophe stripped by `_TITLE_TOKEN_RE`), which doesn't substring-match the bare `superman` in source bodies. Plurals had the symmetric problem: `girlfriends` query token vs `girlfriend` source mentions.

`_stem_token_for_match` is a one-line lite stemmer: trailing `s` strip on tokens of length > 4 that don't end in `ss`. Conservative — short tokens (`"is"`, `"as"`, `"us"`) keep their `s`; `"miss"`/`"class"`/`"chess"` are skipped. Both query tokens & document tokens pass through it before set comparison & body counting; matches in either form score.

This is a soft normalization step BEFORE the hard hash. `question_hash` itself does NOT stem — it canonicalizes by article-strip + trailing-punct-strip, which is enough for the equivalence-class layer. Stemming lives only in the retrieval ranker, where over-recall is preferable to under-recall (filtering happens later).

13.4.6 Per-source context cap — no single doc monopolizes the budget

Fox 2026-04-29: a query for `"who is batman?"` returned 8 sources but the LLM cited only 2 in its answer. Cause: `List_of_Batman_comics` (an 80 KB+ bibliography) was the top FTS5 hit & consumed the entire 60 KB `max_context_chars` at hit #1, leaving every subsequent source dropped with `char_budget <= 0`.

The fix caps each top-K hit at `max_context_chars / top_k` chars before respecting the global budget. With `top_k=8` and `max_context_chars=60000`, each source ships at most 7,500 chars. Bibliographies still appear in the source list but as one slice among many; the actual character article (`Batman`), the franchise page, the film article, the rogue's gallery, etc., all reach the LLM. Multi-source diversity restored without raising the global budget.

```
before:  8 sources retrieved, 1 in context (List_of_Batman_comics ate 60 KB)
after:   8 sources retrieved, 8 in context (each capped at 7,500 chars)
```

13.4.7 Worked example: dinosaurs in Jurassic Park

Query: `"what dinosaurs where in jurassic park film?"`. After stopword + stem: `{dinosaur, jurassic, park, film}`. 4 tokens, so `title_breadth = N - 1 = 3`.

```
title-overlap accept (3 of 4 stems in title):
    Jurassic Park (film)         {jurassic, park, film}
    Jurassic Park (film score)   {jurassic, park, film}
    Jurassic Park III (film score) {jurassic, park, film}

body-density accept (≥ 3 of 4 distinct tokens in body, total ≥ 3):
    Jurassic Park (franchise)
    List of Jurassic Park characters
    Jurassic Park: Operation Genesis
    Jurassic Park video games
    The Lost World: Jurassic Park
```

8-source context, each ≤ 7,500 chars after wikitext-base strip. The model produced 4 quoted spans (Brachiosaurus, Velociraptor, T-rex, Dilophosaurus); 3 verified verbatim, 1 contained the model's own `[...]` elision marker & failed substring match. Final classification: HYBRID / quote / 3-of-4. The unverified span surfaces in `aborist inspect`, where an operator can decide whether the elision is benign or worth re-asking under a tighter prompt.

13.5 Federation: Mesh Layer (Off by Default)

Two peers that ingest our same dump compute bit-identical document_roots & cache_keys. Our mesh layer in aborist is our wire-and-trust scaffolding that lets them gossip those identities, dedup answers, & cleanly distrust an evicted member.

Cryptography (via *cryptography* library):

- **Ed25519** signs every membership mutation & every gossip envelope.
- **X25519 ECDH** wraps each epoch's symmetric mesh secret to every current member's DH pubkey. HKDF-derives a 32-byte AEAD key from each shared secret.
- **ChaCha20-Poly1305** authenticates payloads & per-member secret wraps; bodies optionally encrypt under the per-epoch shared secret with envelope metadata as AEAD additional data.

Membership state is per-epoch. Adding a member, kicking a member, or rotating our secret bumps our epoch & writes an audit event. Eviction works by rotating to a new epoch whose envelope omits our kicked member: their prior signatures stay verifiable forever (their old roster row is preserved), but they have no entry in our new envelope, so any AEAD-protected gossip from epoch+1 onward is opaque to them. Re-admission requires a fresh add operation by an admin.

Off by default: a *mesh.enabled* flag in our meta table gates everything. Fresh installs report *enabled: false* & no network code paths execute until our user runs *aborist mesh init* & *aborist mesh enable*.

13.5.1 Wire protocol

The HTTP gossip wire ships in `aborist/mesh/wire.py`. A peer runs `aborist mesh serve --host 0.0.0.0 --port 8400` to bind a stdlib `ThreadingHTTPServer`; another peer's client (`aborist mesh sync --peer http://...` and `aborist mesh pull --peer http://... --root <hex>`) speaks five message types, all wrapped in a single signed envelope:

```
ANNOUNCE_ROOT          document_root + source_uri + chunking/canonicalization/schema versions
ANNOUNCE_DERIVATION    core_root <- surface_roots(s), distiller_id, proof_blob hash
ANNOUNCE_PROVIDENCE    cache_key + audit_mode + answer_hash + verifier counts
```

```
ANNOUNCE_FALSIFICATION cache_key + reason + witness signature
REQUEST_BODY/DELIVER_BODY  body pull on local miss + per-chunk leaf hashes for Merkle re-derivation client-side
```

Receivers verify the Ed25519 signature against the sender's `sign_pub` looked up in `mesh_roster` at the envelope's `epoch_id`; non-members of that epoch produce no valid signature & their messages are silently dropped (HTTP 401, no audit event). Each accepted `ANNOUNCE_*` writes one `mesh_received` event into the local audit chain whose body is the full signed envelope, preserving non-repudiation. `REQUEST_BODY` triggers a `DELIVER_BODY` carrying the document text + per-chunk leaf hashes; the client re-derives the Merkle root from those leaves & refuses to insert any body whose leaves don't reconstruct the requested root.

13.5.2 Chain-of-claims tracking & fork detection

Per the protocol contract, audit chains across peers cannot "merge" in the literal sense — each peer's chain is a strict-order log of its own actions — but a receiver can detect when a sender broadcasts inconsistent histories. Each gossip envelope carries:

- `prev_event_hash` — the SENDER's previous event_hash they broadcast (or null for the first message);
- `event_hash = sha256(prev_event_hash || canonical_envelope_bytes)` — the sender's deterministic chain-of-claims hash for THIS message.

The receiver's local `mesh_peer_chains` table tracks, per peer, their last-known `event_hash` & sequence number. On accept: verify Ed25519 signature → recompute `event_hash` from canonical bytes & assert equality with the claimed value (400 on mismatch) → look up the sender's prior `event_hash` & assert equality with the claimed `prev_event_hash` (409 Conflict, "fork detected", on mismatch) → write the `mesh_received` audit event & advance the per-peer chain row. The first envelope a receiver sees from any peer is accepted with any `prev_event_hash` value because no prior is on file.

Catchup (peer sends multiple events to fill a gap) is deferred to a future commit; v1 raises a 409 on missed events & relies on the operator/agent to retry. The wire-level fork detection composes with the local audit chain: a tampered DB locally fails the `event_hash` re-hash chain at ingest time, & a forked sender fails the `prev_event_hash` chain at the first received message after the fork.

13.5.3 Optional AEAD body encryption

Signing the envelope guarantees integrity; the body is cleartext by default for backward compatibility. Senders can opt body content into AEAD encryption under the per-epoch shared secret: at construction time, body content is canonicalized to JSON bytes, ChaCha20-Poly1305 encrypted with a fresh 12-byte nonce & `f"{type}|{sender_id}|{epoch_id}|{ts}"` as additional authenticated data, & the resulting `{nonce_b64, ct_b64}` lands in a new `encrypted_body` envelope field while `body` becomes a placeholder. Both fields participate in `canonical_bytes()`, so the Ed25519 signature binds the ciphertext — a MITM cannot swap encrypted_body without invalidating the envelope.

Receivers detect encryption by the presence of `encrypted_body`, recover the epoch shared secret via the existing `mesh.state.recover_epoch_secret` (audit-chain → rotator → ECDH-unwrap-own-slot, no new sender-side cache needed), AEAD-decrypt, & continue normal handler logic with the recovered cleartext body. A receiver missing the epoch secret (evicted member, never enrolled) returns 401 — confidentiality failure is indistinguishable from auth failure at this layer. Mixed-mode peers interoperate: cleartext envelopes still verify byte-identically to the v1 wire format because `encrypted_body=null` is dropped from the canonical form before signing.

13.5.4 What converges across peers

Three classes of derivative reconcile across peers:

- **Deterministic distillers** (e.g., first-sentence cores): same src_root → same core_root, on every machine. Shareable as-is via the existing *derivations.proof_blob*.

- **Corpus-statistical distillers** (e.g., TF-IDF keywords): convergent only under shared corpus scope. Requires a *scope_root* commitment so peers explicitly agree on which corpus statistics underlie our cores.
- **LLM-derived answers**: non-convergent (serving non-determinism), but cryptographically witnessable. Same cache_key from many peers + the same *answer_hash = sha256(answer_text)* collapses to one row with many witnesses; divergent answers stay as multi-witness candidates, each Merkle-bound to a verifiable context_root.

The full operator runbook for a two-host federation lives in `docs/mesh-deploy.md` & the protocol contract (with diagrams) in `docs/mesh.md` in the aborist repo.

13.6 Corpus Snapshots: One Hash Names Our Forest

A *snapshot_root* is *MerkleTree.build([sorted DISTINCT document_roots]).root*: one 32-byte hash naming our content-addressed forest at a point in time. Two peers that ingested our same dump compute bit-identical snapshot_roots, so cross-machine "are we synced?" becomes a single hash comparison instead of doc-by-doc gossip.

Snapshot creation pins our root into our audit chain & records *(snapshot_root, taken_at, audit_event_hash, doc_count, parent_snapshot, reason)*. Snapshots auto-link to our most recent prior snapshot, giving a chain for free. The same *snapshot_root* doubles as our TF-IDF *scope_root*, so derivative reconciliation in 13.5 maps directly onto snapshot agreement.

13.7 Cross-Machine Convergence: A Worked Example

Our 2010-11 enwiki corpus, ingested on a single workstation in 1h 41m (4-way sharded), produces:

```
snapshot_root = 43797e46605de08dbab06cdcaf5be7ad78243b193c56f8580200dee6bcc7e1b9
doc_count     = 3,468,134 (deduplicated)
storage       = 38 GB across 4 shards
chunks        = 6,235,672
edges         = 90,593,523
audit_events  = 3,468,308 (chain intact, 30/30 random Merkle proofs verified)
```

A query against this corpus for "tell me plot of alien film" returned a cache hit:

```
document_root = 896c27f8d9de8e0fc8c9b337274e794bea4206dccd50b9654b34833d9d29f005
document_uri  = https://en.wikipedia.org/wiki/Alien_(film)
context_root  = 896c27f8...f005   (single source, equals document_root)
cache_key     = 1aa20457e980f9e564641b7c3a86edbc90bd9fb31d6e461e9da2fdc4082ddda1
answer        = (plot summary verbatim from the article)
```

Every one of those four hashes is bit-identical on any machine that ingests our same *enwiki-20101011-pages-articles.xml.bz2* (verifiable via our *md5sums.txt*) with our same aborist commit. Hand *896c27f8…f005* to any peer; if their corpus matches ours, they compute our same hash, with mathematical certainty. No trust required, just shared inputs.

What does NOT converge across two machines:

- *audit_events.event_hash* chain (event order is per-instance witness, not content)
- *documents.ingest_ts* & *chunks.chunk_id* (per-instance metadata)
- *answer_text* under fresh LLM inference (Hermes serving carries tiny non-determinism)

Everything in our cache key dimensions converges. Everything that observes our corpus's processing diverges. That separation is our admissibility property: one peer's witness of "what was in our forest at time T" is bit-equal to every other peer's witness of our same forest, while one peer's audit chain proves *that peer* witnessed our forest at our times *they* recorded.

13.8 Audit Mode: Source vs. Emergence

A Merkle proof binds an answer's *context* to a content-addressed root. It does not bind the *answer* to that context. A model handed a chunk of episode plot summaries can correctly recite the entity names that appear in the chunk, then complete the response with character bios drawn from training data — names that match, claims that emerge. Both layers are real. Conflating them under one "STRICT" label overclaims the proof.

v9.8 names the trichotomy (Section 13.1 inherits it from Merkle-AGI's audit contract): STRICT, HYBRID, VISUAL. Aborist decides which label applies *per answer*, by post-LLM faithfulness check, never as a default. The substrate's "VISUAL" label was named for neural-network proof systems where it meant "FOR-style visualization, no formal guarantee attached." In a RAG layer that semantic reads as *no recoverable proof of grounding*, so aborist persists the label as **UNGROUNDED**:

```
STRICT      every evidence unit in answer verifies verbatim against context
HYBRID      some claims source-grounded, some emerged from training
UNGROUNDED  no evidence, or none verifies — purely emergent
```

13.8.1 Layered verifier

aborist/qa/verify.py runs four strategies in sequence. The first to find evidence classifies the answer; later strategies don't run. Each strategy is a lexical operation under norm-v1 canonicalization (NFC, collapsed whitespace) plus case-folding — substring tests for strategies 1-3, token-coverage for strategy 4. The soft-hash boundary from CLAUDE.md applies: embeddings train rankers, hashes carry proofs, & we never mix the two. `verifier_method` on the providence record names which path fired so the audit chain stays diagnostic.

Strategy 1: quote. The system prompt instructs the model to wrap every factual claim in a verbatim double-quoted span from a source. The verifier locates every double-quote character in the answer & pairs them sequentially: 1st & 2nd, 3rd & 4th, & so on. Each pair brackets one quoted span; text between consecutive pairs is the model's framing prose & is not captured. Pure-regex pairing fails on adjacent quote pairs like `"title" prose "quote"` — the regex captures `prose` as a phantom span because every `"` looks like both an opener & a closer to it. Sequential pairing eliminates the artifact.

Strategy 2: span. When the model declines to quote but writes lines verbatim from source, bullet & sentence units from the answer are substring-tested against context. Catches paraphrased-into-prose answers where every claim is grounded but the formatting drifted from explicit quote marks.

Strategy 3: entity. When neither quote nor span lands evidence, multi-word capitalized phrases (Carrie-Anne Moss, Thomas A. Anderson, Agent Smith) are extracted from the answer & substring-tested. Catches the Wikipedia-infobox case: source has structured `[[Keanu Reeves]] [[Laurence Fishburne]]` markup; model paraphrases into prose; spans diverge but every named entity is intact & grounded. Entity-existence in source is, however, weaker than claim-level evidence — the model could correctly name entities while making *claims around them* that came from training. Aborist gates the entity path with a policy:

```
strict     all entities verify → STRICT (legacy; overclaims TMNT-style cases)
hybrid     any entity verifies → HYBRID (honest cap)
drop       skip entity path entirely → UNGROUNDED (most conservative)
proximity  STRICT only if N=3 verified entities cluster within W=300 chars
           in source. Otherwise demoted to HYBRID/UNGROUNDED.
```

The default policy is **proximity**. Entity-cluster density distinguishes "source documents these entities as a group" (cast list, infobox, roster) from "source mentions them incidentally in scattered prose." A Wikipedia infobox passes the cluster test & promotes; a TMNT episode plot summary that names the four turtles & Splinter across separate sentences fails it & demotes. The policy lives in `DEFAULT_QUERY_POLICY["entity_policy"]` so any change folds into `governance_policy_hash` & invalidates prior cache cleanly.

Strategy 4: paraphrase. The span strategy's secondary path. When a sentence-shaped span fails substring match in strategy 2, a token-coverage probe runs on the same span: tokens of length ≥ 4 chars are extracted (less per-token punctuation strip so `wayne,` lines up with bare `wayne`), filtered against an English stopword

set (from, with, have, would, which, where, their, etc. — high-frequency fillers that match almost any English text), and the fraction of remaining content tokens present anywhere in the normalized base context is computed. The span is accepted as paraphrase-verified when coverage clears DEFAULT_PARAPHRASE_COVERAGE (default 0.85) with at least DEFAULT_PARAPHRASE_MIN_TOKENS (default 4) content tokens. The label verifier_method='paraphrase' distinguishes records where any soft-verified items contributed from records where every match was lexical-verbatim. Paraphrase fires only on prose-shaped spans, not on lists of proper nouns — a span needs at least 2 lowercase content tokens (_is_prose_span) to qualify, so a list like "Keanu Reeves, Laurence Fishburne, Carrie-Anne Moss" continues through to the entity strategy where proximity policy can tight-cluster check.

The stopword filter is calibration choice, not threshold relaxation. Including stopwords in the denominator inflates coverage scores for cases where topical content is missing — a fabrication like "Batman is the alias of Bruce Wayne, a wealthy businessman who resides in Gotham City" (where wealthy / businessman / resides aren't in source) would score ~0.67 raw but should NOT promote. Filtering the stopwords drops the denominator to topical tokens only; the missing topical tokens dominate the score & the span correctly stays UNGROUNDED. Lowering the threshold itself (e.g. 0.85 → 0.67) would have promoted the fabrication; tightening the signal preserves discrimination.

Token-coverage is a soft signal: it answers "the same content tokens appear in source," not "the answer is a byte-equivalent transcription." The hard chain still records the trichotomy classification, but auditors reading STRICT can use verifier_method to tell verbatim-from-source-text from same-tokens-different-sequence. Quote strategy (strategy 1) deliberately does NOT receive paraphrase fallback — the double-quote convention asserts verbatim citation, & paraphrasing-then-quoting is a model error to flag, not auto-promote.

13.8.2 Wikitext base prose

Aborist stores raw MediaWiki wikitext in chunk content so the link graph & original markup are recoverable from any page. The LLM produces clean prose. Without normalization, every wikilink-carrying source paragraph compares as a different surface form & the verifier wrongly reports UNGROUNDED on genuine source-grounded quotes — the FF7-characters page was the input that triggered the design (six legitimate quotes flagged as ungrounded because the model wrote Cloud Strife & the source had [[Cloud Strife]]).

aborist/wikitext.py provides to_base(raw), a pure function pinned by BASE_VERSION (wikitext-base-v1). Same wikitext → same prose, forever, until BASE_VERSION is bumped. The algorithm runs mwparserfromhell: drop <ref> tags & namespace-prefixed wikilinks ([[File:...]], [[Image:...]], [[Category:...]]), call strip_code(normalize=True, collapse=True) to reduce surviving markup to display text, then collapse whitespace runs. Optional dependency — installations without mwparserfromhell fall back to identity & both LLM and verifier proceed against raw wikitext (the strip becomes a no-op rather than a hard requirement).

The conversion runs in **two places**, gated by the same policy["base_version"] flag:

1. **Before the LLM call.** aborist/qa/runner.py and aborist/qa/query.py apply to_base() to the assembled context before the user-turn arrives at Hermes. The model never sees [[wikilinks]], {{templates}}, or <ref> tags — it sees the prose a human reader would see. Two consequences: the model can quote source paragraphs verbatim instead of escaping markup it doesn't understand, & Wikipedia chunks ship with ~43% fewer tokens (measured against the FF7-characters chunk). The token reduction compounds across context windows packed with multiple sources.
2. **Inside the verifier.** verify_quotes re-applies to_base() to the context before substring-testing the model's quoted spans. Idempotent — passing already-stripped prose through to_base() returns it unchanged. Same canonicalization on both sides means the verifier compares like-against-like.

policy["base_version"] lives in DEFAULT_POLICY and DEFAULT_QUERY_POLICY so it folds into governance_policy_hash. Bumping BASE_VERSION (e.g. wikitext-base-v1 → v2 after an algorithm change) invalidates every prior cache record's 8-dim key on next lookup; no schema migration, the next ask re-derives against fresh prose.

13.8.3 Emergence is preserved, not suppressed

A HYBRID or UNGROUNDED record stays live in cache. Training-derived knowledge is often correct (the turtles really do wield katanas), just unverifiable against the current corpus. We label honestly so a downstream consumer can route: STRICT answers carry compliance-grade provenance, UNGROUNDED answers carry *the model believes this, no certificate attached*. A model that genuinely refuses ("I don't know based on the provided sources") classifies UNGROUNDED with `verifier_method='none'` & `n_quotes=0` — the most honest answer the system can produce.

Emergence is a corpus-growth signal. Unverified spans persist on each providence record; `aborist emergent --aggregate` ranks them by frequency. Spans the model produces repeatedly that the corpus cannot ground are candidate ingest targets — what training has that ingestion is missing. When the missing source eventually arrives, `aborist reclassify` re-runs the verifier against existing answers without any LLM call, persisting the upgraded label & writing one `providence_reclassify` audit event per changed row. HYBRID promotes to STRICT, UNGROUNDED to HYBRID, & the audit chain remembers the transition. Prior records stay as historical witnesses: *on this date, against that corpus root, the model emerged this much beyond its sources.*

13.8.4 Verifier stays binary; falsifications carry soft signal

The verifier returns evidence units & a classification — that's the entire output surface. No per-quote diagnosis fields, no match-confidence scores, no partial-match indicators on hard verifier output. "Why didn't this ground?" lives on the falsify+reclassify loop where an operator (or an automated process) inspects the unverified spans, decides whether the failure is a corpus gap or a model fabrication, & either ingests new source or quarantines the record. Keeping the binary discipline on the verifier preserves the soft-hash boundary: classification is a hard bit, not a soft score.

The SOURCE/EMERGENCE distinction also clarifies the v9.8 cache_key invariant. Bumping `governance_policy_hash` (e.g. by tightening the system prompt or flipping `entity_policy`) invalidates cache lookups but leaves the source merkle tree, chunks, document_roots, audit chain, & snapshots bit-identical. Q&A records under the old policy stop being reachable; corpus identity is unchanged. No re-ingest required — the 8-dim key partitions cleanly along architectural seams.

13.8.5 Trailing-citation strip

Models trained on cited text often append parentheticals like `(Source: https://...)` or `(citing Wikipedia)` after a verbatim source sentence. Without preprocessing, the verifier substring-tests the prose **plus** the appended citation; the citation isn't in the corpus, so the whole span fails & the answer flags HYBRID or UNGROUNDED. The prose itself was verbatim.

`extract_quotes` and `extract_claim_spans` strip a single trailing parenthetical at end-of-span before substring testing, gated on a conservative regex: the parenthetical contents must start with a citation cue word (`Source:`, `citing`, `see`, `ref`, `reference`, `from`) OR contain a URL. A genuine prose parenthetical like `Pikachu (a Pokémon species) lives in forests` carries no cue word & no URL & stays untouched. The strip recovers verbatim-source spans without weakening the verifier on real disagreements.

13.8.6 Sidecar diagnostics

The verifier's output surface is deliberately small: an audit_mode label, an n_verified count, the unverified spans, & the verifier_method that fired. No per-quote diagnosis fields, no match-confidence scores, no nearest-neighbor traces. Keeping the binary discipline preserves the soft-hash boundary — classification is a hard bit, not a soft score, & enters `governance_policy_hash` cleanly.

But operators still need to ask "*why* didn't this span ground?" when triaging the unverified list. v9.8's answer is a **sidecar**: a read-only diagnostic verb that pulls the same source chunks the verifier saw, runs the same canonicalizations, & classifies each unverified span against the corpus. Sidecars never write to `providence_cache`, never extend `audit_events`, never alter any v9.8 field. They run on demand, observe state, & report. The classification-as-hard-bit invariant survives because sidecars participate in no proof path — they help an operator triage; they do not change what's signed.

The reference sidecar in aborist is `aborist inspect --cache-key X`. Each unverified span lands in one of:

- `verbatim_in_base` — the span IS in the base context. Surfaces a verifier or canonicalization regression so an operator can fix the root cause.
- `verbatim_in_raw_only` — span matches raw wikitext but not the base form. Wikitext-strip edge case worth its own label.
- `trailing_artifact` — a long prefix (≥ 60 chars) matches; the tail is reported separately. Catches model-appended citations that the strip in 13.8.5 didn't reach (different cue words, multi-line, etc.).
- `paraphrase` — high token coverage (≥ 0.85) but not in this sequence. Model rewrote source content; ingest more sources or tune the prompt.
- `partial_paraphrase` — 40-85% token coverage. Mixed sourced/emergent content within a single span.
- `no_overlap` — < 40% coverage. Likely full invention; corpus-growth signal.

The sidecar pattern generalizes beyond *inspect*. Any future tool that wants to surface *why-not-grounded* — embedding-based nearest neighbors, LLM-generated rationales, fuzzy-match scores — belongs in a sidecar. The hard chain stays binary; the soft channel proliferates as the project learns. Sidecars compose cleanly because they all read from the same v9.8 record & contribute nothing back.

13.9 Claim-Lattice-Pointer Mode (CTI / Quote-by-Pointer)

The quote/span/entity/paraphrase verifier in 13.8 audits answers the model wrote in its own words. It catches one defect class cleanly — *did the quoted text appear verbatim in source* — but it cannot catch a related class: **synthetic elision**. A model emitting `"prefix [...] suffix"` produces a span whose halves both verify on their own, glued together by a model-typed ellipsis that flattens our ground truth. The verifier reports STRICT on a frankenquote.

Aborist's response is a second answer mode that makes synthetic elision **structurally impossible**: the model never types the quote string. Instead, the runtime owns extraction, the model owns selection, & the renderer owns display. We call this *Clause Lattice Intelligence* (CTI), or in implementation terms `policy["answer_mode"] = "claim_lattice_pointer"`:

```
raw model output
→ clause extraction by pointer parser
→ claim nodes with evidence pointers
→ deterministic verifier
→ admissibility decision
→ rendered prose with literal source spans
```

The model writes natural prose with bracketed evidence-id tags:

```
Steve Jobs co-founded Apple. [E1]
Steve Wozniak co-founded Apple. [E1,E2]
Ronald Wayne co-founded Apple. [E1]
```

Each `[E\d+]` references an entry in a runtime-built **evidence map** the model saw in the prompt. The runtime carries two ids per evidence object — a short `pointer_id` (`E1`, `E2`, …, sequential within the run) shown to the model, & a `evidence_id` (sha256-derived `E########`, content-addressed) used by the cache, run-DAG, & audit chain:

```
EvidenceObject {
    pointer_id    "E1"                                    # model-facing
    evidence_id   "E1f8e4c2a"                              # run-stable
    source_root   <document_root>
    chunk_root    <chunk leaf hash>
    offset_start  0
    offset_end    <len(span)>
    source_role   primary_answer_source
    text_hash     sha256(span)
```

```
    span          <literal source text>
}
```

Two-layer id discipline is load-bearing. Showing the model the hex form (`E1f8e4c2a`) tokenizes to 4-6 BPE tokens of out-of-distribution noise that nudges Hermes 3 toward DSL/code-completion mode. Showing it `E1` is one BPE token & lands inside the citation-style prose distribution that academic papers, Wikipedia articles, & footnoted text taught the model. Citation style is heavily represented in training; random hex is not. The runtime maps pointer_id → evidence_id at verification time; everything downstream of the verifier sees the content-addressed form.

13.9.1 Why pointer lines, not JSON

JSON is one encoding for *(claim_text, [evidence_ids])* pairs but not the only one — & not the right one for an 8B instruction-tuned model. JSON discipline failures (missing braces, prose preamble before the object, escaping errors) generate spurious one-shot SCHEMA_INVALID verdicts even when the model knew the right answer. Pointer-line stays inside the model's prose distribution: the model writes prose, sometimes with bracketed citations, exactly as it has seen citations written ten thousand times. The parser walks lines, pulls every `[E\d+]` & `[E\d+,E\d+,…]` bracket payload, & returns `(claim_text, pointer_ids)` per line. Lines without a tag get `parse_status = "NO_EVIDENCE_POINTER"` & count toward the denominator so the model can't smuggle unsourced prose past the verifier. Lines with a tag but no surviving claim text after the strip get `parse_status = "SCHEMA_INVALID"`.

13.9.2 Hard verifier — five deterministic checks

`verify_claim_lattice` runs **only** deterministic checks. The hard/soft boundary from 13.8.4 holds — semantic entailment, completeness, predicate compatibility, scope ambiguity, & counterevidence retrieval all stay sidecar territory. The hard checks:

```
1. parser_succeeded            line had a recognized [E\d+] bracket
2. evidence_id_resolves        pointer maps to an entry in the
                               runtime-built evidence map
3. source_role_allowed         resolved entry's source_role is in
                               the policy allowlist
4. no_manual_quotes            zero double-quote characters in claim
                               text — strict, any " violates
5. claim_text_non_empty        the line carried prose around the tag
```

Aggregation matches 13.8's trichotomy: STRICT = ≥1 verified pair AND zero violations of any kind; HYBRID = some pairs verified, some failed; UNGROUNDED = no verified pairs. Per-claim statuses sharpen the diagnosis without breaking the binary discipline: `EVIDENCE_LINKED`, `EVIDENCE_LINKED_PARTIAL`, `UNKNOWN_EVIDENCE_ID`, `SOURCE_ROLE_BLOCKED`, `MANUAL_QUOTE_VIOLATION`, `NO_EVIDENCE_POINTER`, `SCHEMA_INVALID`.

The strict no-quotes rule deserves its own line. Even a 4-char quoted span (`"hi"`) is a model-asserted verbatim citation the runtime didn't authorize. Pointer mode forbids them entirely — the runtime owns quotes via the renderer, not the model. Forcing this property cleanly is the entire architectural contribution: *the model that cannot type a quote cannot synthesize an elision in one*.

13.9.3 Spotlight render — cited evidence, one window per claim

The renderer interpolates literal source spans from the evidence map at display time. A naive truncation (`span[:200]`) shows the same article-intro sentence under every claim when the model lazy-anchors at the topic id. Aborist instead extracts content tokens from each claim text (≥4-char alphanumeric tokens, stopword-filtered, longest-first) & finds the first match in the cited span. The displayed window centers on that match; if no token matches, the renderer falls back to the leading window with a trailing ellipsis. *Same evidence pointer, different displayed sentence per claim* — anchored on what the claim is actually about.

13.9.4 Per-chunk granularity & query-relevance ordering

A whole-document evidence object lets a small model lazy-anchor: cite `[E1]` (the topic article) for every claim & let the renderer figure out the rest. The fix is structural — each retrieved source contributes one `EvidenceObject` per chunk, not one per source. The model sees 30+ short paragraph-scale spans tagged `E1`…`E30` & must pick the specific paragraph that supports each claim, or omit the claim.

Within each source, chunks are ranked by query-token overlap before `pointer_id` assignment:

```
sort key:
    -distinct_query_tokens_present     # primary: more topic words wins
    -total_mentions                    # tiebreaker: density wins
    +chunk_idx                         # final tiebreaker: doc order
```

So `E1` of a source is the chunk most likely to textually support the question, not the chunk that happens to be first in document order. This is a soft signal — token overlap, never enters the proof path — but it shapes the model's anchoring without giving it semantic discretion. Hermes-3-8B's lazy-anchor habit (cite the first thing you see) lands on a relevant chunk by accident.

13.9.5 9-stage CTI run-DAG

The per-run Merkle-DAG (`run_dag_root`) grows from 7 stages to 9 in pointer mode:

```
quote mode (default, 7):
    question / retrieval / context / prompt / answer /
    verify / final_label

pointer mode (9):
    question / retrieval / evidence_map / prompt / raw_answer /
    parsed_claim_lattice / verify / render / final_label
```

The new stages commit each provenance step independently:

- `evidence_map` — Merkle root over sorted `evidence_ids`, order-independent so two retrieval runs that returned the same chunks in different orders produce the same root.
- `raw_answer` — sha256 of the model's literal pointer-line output.
- `parsed_claim_lattice` — sha256 of the canonical-JSON of the parsed claim nodes (claim_text + content-addressed evidence_ids per claim). Reordering claims changes the hash.
- `render` — sha256 of the rendered prose with literal spans interpolated.

`context` drops out — the context IS the evidence map. `answer` splits into raw / parsed / render so an auditor can reconstruct each transformation step from the persisted blob. Quote mode keeps the original 7-stage shape so pre-G0 `run_dag_root` values stay valid.

13.9.6 No iterative repair — one-shot discipline

Pointer mode skips the mechanical-repair + reprompt-feedback loop that 13.8 reserves for quote mode. The benchmark contract is *zero-shot ideal, one-shot maximum, no repeated retries until success*. A run may fail honestly. A run may self-label uncertainty. A run may downgrade itself. A run may not keep trying until it passes. UNGROUNDED beats hallucinated success.

The hard discipline carries a measurable cost: small models drop the bracket protocol on perhaps half of cold runs. The lever is the prompt, specifically a one-shot worked example showing the format end-to-end. With the example, JP dinosaurs benchmark went 0/13 → 10/11 → 16/17 verified across the iterations that hardened the pipeline. Without the example, runs were near-random in protocol compliance.

13.9.7 What pointer mode does NOT prove

The pointer-mode verifier proves *evidence-linked-and-role-OK*. It does **not** prove the cited evidence textually entails the claim. A model citing `[E1]` for "Brachiosaurus appears in the film" passes the hard checks if E1 exists, has an allowed source role, & the claim has no manual quote — even if E1's text never says "Brachiosaurus." That gap is by design. Entailment requires either an NLI model (soft signal) or LLM-as-judge (Hermes round-trip) — both of which would re-introduce the soft/hard boundary leak the architecture is built to prevent. The honest report is "this claim is evidence-linked & quote-safe; semantic entailment is not yet hard-verified," and the spotlight renderer makes the limit visible: when the cited span doesn't contain the claim's terms, the displayed excerpt falls back to the leading window — an auditor reading 16/17 EVIDENCE_LINKED + leading-window excerpts immediately sees "model lazy-anchored; the cited chunk doesn't textually support the claim."

The reverse-RAG / claim-lattice / merkle-providence stack now covers four layers cleanly: provenance commitment (Merkle-AGI), source linkage (Reverse-RAG), claim decomposition (CTI), & quote-safe rendering (spotlight). The next layer — semantic warrant — is intentionally deferred until a deterministic-enough entailment substrate exists to enter the proof path without staining the soft/hard boundary.

14. Related Work

Graphify (*pip install graphify*) builds navigable knowledge graphs from local folders: code in 13 languages, PDFs, images, & markdown. One command produces an Obsidian vault, a wiki, & a graph queryable in plain English. Graphify's authors report significant token reductions for graph-traversal queries versus raw file reads. Our work complements Graphify: where Graphify builds graphs over static local folders, Merkle Providence describes verifiable answer chains over live web documents & git-versioned codebases, with cryptographic provenance for every cached result. No prototype of our integration exists yet.

Git / Mercurial object model. Both git & Mercurial implement content-addressed Merkle DAGs natively. Git's commit object hashes our entire tree; Merkle Providence inherits this as a free cache boundary signal. Our work makes this implicit property explicit & useful for ML inference caching.

Traditional RAG (Lewis et al., 2020). Retrieval Augmented Generation introduced server-side vector retrieval for language models. Our Reverse RAG paper documented client-side inversion of this pipeline. Merkle Providence extends Reverse RAG with verifiable caching, eliminating our repetition tax that RAG systems do not address.

Zero-Knowledge Proofs. Merkle membership proofs constitute a subset of ZKP: proving a leaf exists in a tree without revealing other leaves. Our work applies this primitive to ML answer provenance without requiring full ZKP apparatus (zk-SNARKs, Bulletproofs). We trade completeness for simplicity & speed.

Acknowledgements

Our Merkle cache idea emerged from a conversation on X Spaces. We thank **@AR_Milne** & **@without_rulers** for a discussion that sparked the core insight: that a Merkle tree over document content could serve as a fast, verifiable cache layer for machine learning inference. Ideas born in public conversation belong to our commons.

Citation

```
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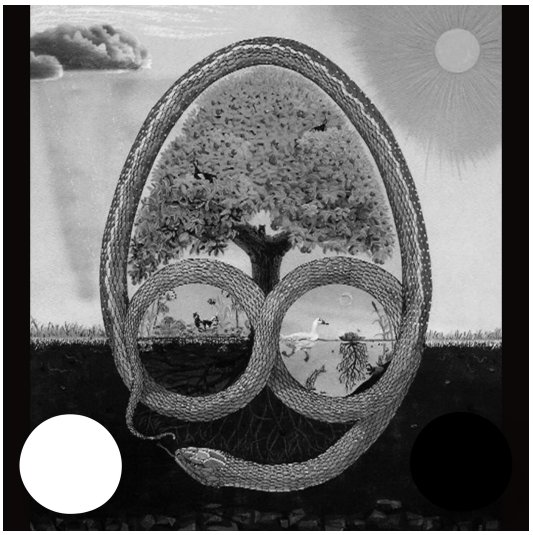

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Corresponding Source. Regardless of what server hosts the Corresponding Source, you remain obligated to ensure that it is available for as long as needed to satisfy these requirements.

e) Convey the object code using peer-to-peer transmission, provided you inform other peers where the object code and Corresponding Source of the work are being offered to the general public at no charge under subsection 6d.

A separable portion of the object code, whose source code is excluded from the Corresponding Source as a System Library, need not be included in conveying the object code work.

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